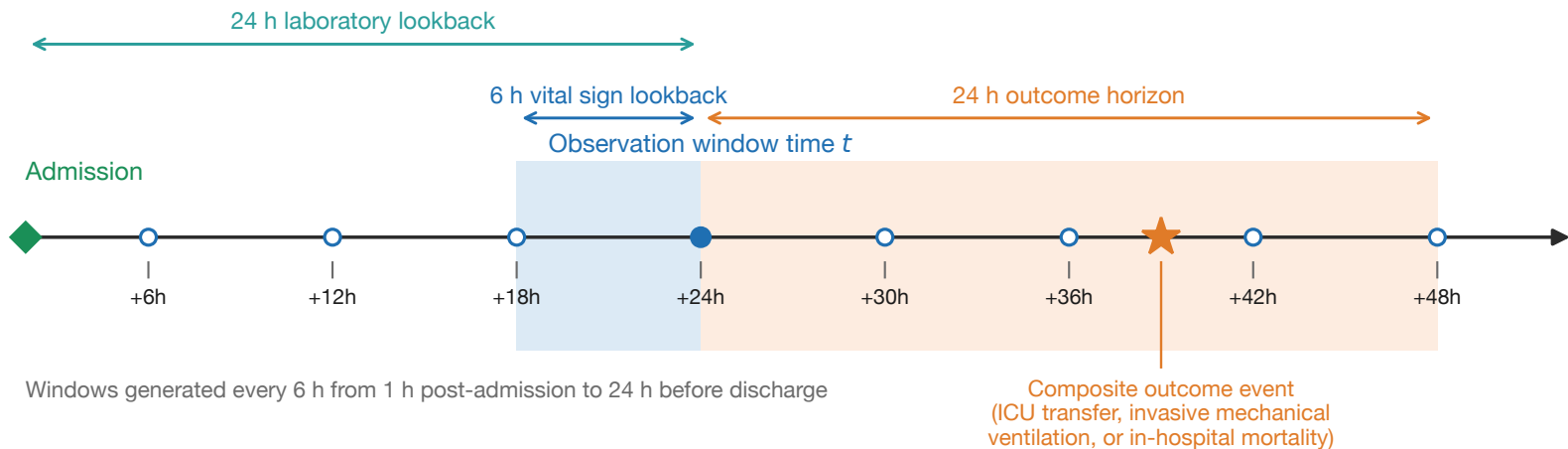


a



b

Model input feature domains

Vital signs	28
Heart rate, SpO ₂ , BP, respiratory rate, temperature, GCS	
Missingness indicators	20
Absence flags for 12 laboratory and 6 vital sign variables, plus 2 counts	
Laboratory values	12
Lactate, creatinine, WBC, INR, bilirubin and others	
Temporal / demographics	12
Hours since admission; age; sex; admission type	
mNEWS2 subscores	8
Computed from aggregated vitals (RCP 2017 specification)	
Medications	3
Vasopressors, antibiotics, anticoagulants	

Total features 83

Domain sizes as used in the leave-one-domain-out ablation analysis (Table 5)

c

Internal validation performance

AUROC (bssentinel)	0.758 (95% CI 0.753–0.763)
AUROC (mNEWS2)	0.568
AUPRC	0.102
Brier score (calibrated)	0.018
At model operating score 0.31 (raw, uncalibrated)	
Sensitivity	90.0%
Specificity	37.9%
False-positive rate (1 – specificity)	62.1%
PPV	2.8%
NPV	99.5%

Alerts per monitored patient-day 2.51

Five-fold patient-stratified cross-validation; 1,334,773 windows; 26,201 events (1.96%).
De-duplicated into alert episodes: 0.52 per monitored patient-day.